Clément Vurpillot

Data Engineering | Machine Learning | MLOps

Final-year Data & AI engineering student at ESILV – 6-month internship or VIE from February 2027

Houilles (Paris area), France | clement.vurpillot@gmail.com | +33 7 83 41 99 87
linkedin.com/in/clement-vurpillot | github.com/Oradixx | French citizen

SUMMARY

Engineering student (Data & Artificial Intelligence major) seeking a final-year internship or VIE in Data Engineering, Machine Learning or MLOps. Hands-on across the full chain: automated ETL pipelines, SQL and NoSQL databases, machine learning (ML) and LLM applications (RAG), REST APIs, containerised deployment and CI/CD on the cloud. Used to explaining results to non-data teams and to working in English.

EDUCATION

ESILV, Graduate School of Engineering, Paris La Défense 09/2022 – 08/2027
Engineering degree (CTI-accredited, Master's level, 300 ECTS), Data & Artificial Intelligence major.

- Coursework: Deep Learning, MLOps, NLP, LLM & Generative AI, Big Data (Apache Spark), NoSQL Databases, Data Mining.

RTU, Riga Technical University, Latvia 01/2025 – 06/2025
International semester in data science and AI, taught entirely in English.

PROFESSIONAL EXPERIENCE

CERTAM Rouen, France
Data & AI Engineering Intern 04/2026 – 08/2026

- Built an **end-to-end IoT ETL pipeline** for 1 Hz vessel telemetry: J1939 CAN bus → AWS S3 (Parquet) → async Python → PostgreSQL/TimescaleDB.
- Trained a Random Forest rebuilding particulate readings on sensor failure ($\mathbf{R^2}$ **up to 0.96**), explained with SHAP and presented to the test engineers.
- Tuned slow dashboard queries with SQL materialised views: load time cut from **3 hours to 30 seconds**.
- Set up real-time Grafana alerting on a hardened Linux VPS: the pressure-drop alert caught a filter clogging **before engine failure**.

Diduenjoy Paris, France
Data Labeling & Quality Supervisor (internship, then freelance in 08/2025) 07/2024 – 10/2024

- Ran **weekly data quality audits** and evaluated production NLP models (precision / recall); supervised a team of 5 interns under a 72-hour SLA.

PROJECTS

NFL Big Data Bowl, Paris Side Quest (NFL × AWS hackathon): 2nd place *Python, Kiro (AWS), Mistral API*
Team of 3, main contributor 10/2026

- Analysed NFL Next Gen Stats tracking and PFF data (8 weeks of the 2021 season) on "chip" blocks: an in-line tight end **cut pressure from the chipped rusher by 26 %**; shipped a web demo with an LLM-generated coach briefing.

France Grid Forecast, personal project *Python, dbt, DuckDB, Scikit-Learn, GitHub Actions*

- **Daily automated pipeline** forecasting French electricity demand: dbt models on DuckDB with **19 data tests**, gradient boosting model, forecasts published to a live dashboard only if the quality gate passes.

NLP analysis of 24,000 insurance reviews *Hugging Face, BERT, Llama-3, RAG*

- **RAG** summarisation pipeline and zero-shot topic detection; switched from Flan-T5 to a **locally deployed Llama-3 (8B)** after error analysis showed hallucinations; TF-IDF baseline benchmarked against embeddings and BERT.

CryptoIndexLab, containerised microservices (team of 3) *FastAPI, Docker, PostgreSQL, Nginx, AWS*

- Two FastAPI REST APIs behind an Nginx load balancer on isolated Docker networks; CI/CD with a **blocking Trivy security scan**, deployed on AWS.

SKILLS

- **Machine Learning & AI:** Scikit-Learn, TensorFlow, Hugging Face, BERT, deep learning, NLP, LLM, RAG, SHAP.
- **Data Engineering:** Python (Pandas, NumPy), SQL, ETL pipelines, PostgreSQL, TimescaleDB, DuckDB, dbt, Apache Spark (PySpark), Parquet, MongoDB, Cassandra.
- **Cloud & MLOps:** AWS (S3, Lightsail), GCP (GKE), Docker, Kubernetes, GitHub Actions, CI/CD, FastAPI / REST APIs, Linux, Git, Grafana.
- **Languages & certifications:** French (native), English C1 (TOEFL ITP 633/677), German A2. Scrum & Agile methods, IBM Data Fundamentals.